

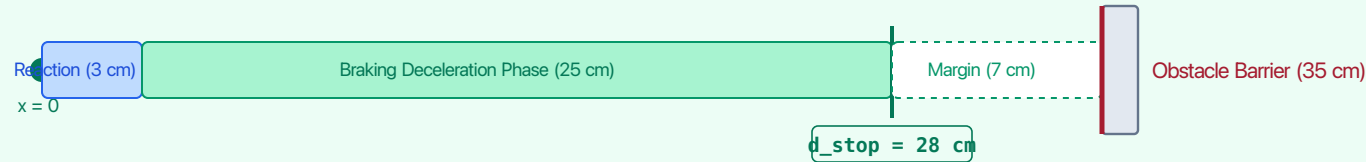
DYNAMIC STOPPING DISTANCE PHYSICS (d_stop)

Translating milliseconds of computational delay into physical centimeters of collision hazard: $d_{stop} = v \cdot \Delta t_{delay} + v^2 / (2 a_{max})$

Governing Physics: Total Stopping Distance = Reaction Travel ($v \cdot \Delta t$) + Braking Travel ($v^2 / 2a_{max}$) | Nominal Parameters: $v = 1.0\text{ m/s}$, $a_{max} = 2.0\text{ m/s}^2$, Clearance Barrier = 35 cm

SCENARIO A · NOMINAL P50 LATENCY ($t_{delay} = 30\text{ ms}$)

VERDICT: SAFE STOP (+7 cm margin)



SCENARIO B · P99 LATENCY TAIL SPIKE ($t_{delay} = 230\text{ ms} \cdot \text{OS Jitter} + \text{UMA Contention}$)

VERDICT: SEVERE COLLISION (13 cm breach)



The Invariant Enforcement Mandate:

Because an untrusted foundation model on the MPU cannot guarantee bounded latency tails under UMA memory contention, the dedicated 1 kHz MCU must continuously